Tyler Bibus

tylerbibus@hotmail.com | 651-210-3421 | tyler-bibus.github.io/personal

Professional Summary

Computer Engineering student at Iowa State University (GPA 3.9) graduating December 2026, seeking full-time roles in hardware design and computer engineering. Experience spans processor and FPGA design, ML hardware acceleration, 5G networking, and full-stack software development, with a record of owning projects end to end.

Experience

Software Developer Intern — Boisei Labs, State College, PA May 2026 – Present

- Sole developer of a full-stack React Native application (under NDA), owning architecture, backend, frontend, and testing end to end and delivering a near-production app within the internship term.
- Designed, tested, and deployed custom AI agent workflows to accelerate development, integrating them into the build and testing process.
- Developed features for a production React web application as part of the engineering team.

Teaching Assistant, CPRE 3810 (Computer Architecture) — Iowa State University, Ames, IA Aug 2025 – May 2026

- Led lab sections for 46 students, supporting debugging and design work in computer architecture.
- Supported the course's migration from MIPS to RISC-V, updating documentation and the student testing framework.

Undergraduate Researcher — ARA Wireless, Ames, IA Jan 2025 – Aug 2025

- Implemented outdoor 5G networking under a graduate mentor, focused on bringing high-speed, reliable internet to rural areas.
- Adapted existing 5G software for outdoor amplifier use, programming software-defined radio (SDR) output pins in C and C++ to achieve a 48 dB signal improvement.
- Tested and deployed Mender for remote, scheduled software updates across tower sites, eliminating manual tower climbs for maintenance.

Undergraduate Research Programmer — CSAFE, Ames, IA Oct 2023 – May 2024

- Collaborated with a remote research team on EviHunter, an Android digital forensics tool.
- Built a web crawler that scraped several hundred Android APK files and metadata for the project database.

CAD Technician — Americom, Eagan, MN Aug 2021 – Aug 2023

- Designed and drafted 150+ AV, network cabling, and security drawings in AutoCAD LT, collaborating with sales and design teams to meet installer and client standards.

Education

Bachelor of Science in Computer Engineering, Minor in Music Expected Dec 2026

Iowa State University, Ames, IA | GPA: 3.9

- President's Award (perfect 4.0 for two consecutive semesters, Fall 2024 onward); 4× Dean's List recipient.

Projects

AI Hardware Accelerator (FPGA) — CPRE 5870, Hardware Design for Machine Learning Fall 2025

- Built an FPGA-based hardware accelerator for an image classification model.
- Benchmarked fast multiplier architectures across quantization sizes and presented findings tied to related research.

MIPS Processor (VHDL) — Single-cycle and pipelined implementations Spring 2025

- Designed a single-cycle processor with 100% MIPS ISA instruction coverage, then developed software- and hardware-scheduled pipelined versions with significant performance gains.

Goods and Service Finder — Android app with Spring Boot backend Spring 2025

- Built an Android app on RESTful APIs with fluid, performant UI/UX using XML and fragment layouts.

Activities

ChipForge (ASIC design club — industry-standard design, testing, and fabrication practices) | Iowa State Symphonic Orchestra (bass) | Food At First (weekly food shelf volunteer)

Skills

Languages: C, C++, Java, Python, JavaScript, VHDL, Verilog, MIPS Assembly, HTML
Hardware & EDA: FPGA design, ASIC design, QuestaSim, ModelSim, LTspice, AutoCAD LT
Software & Frameworks: React, React Native, Spring Boot, Android Studio, TensorFlow, Docker, Kubernetes, CI/CD
Machine Learning: Model training, quantization, hardware acceleration, AI agent workflows